

Assignment 2: Classification Decision Tree using the ID3 Algorithm

Dataset: Titanic-Dataset

Objective: To understand the implementation of the ID3 (Iterative Dichotomiser 3) algorithm for building a classification decision tree and evaluate its performance on a dataset.

Task Instructions:

❖ Dataset Selection and Preprocessing

- Load the dataset and perform basic data preprocessing, including handling missing values.
- Split the dataset into training and testing sets (80:20 split).

❖ Implementing ID3

- Implement the ID3 algorithm from a Python library like scikit-learn to construct a decision tree classifier.
- Key steps:
 - Calculate entropy for the target class.
 - Calculate information gain for each feature.
 - Select the feature with the highest information gain for splitting.
 - Recursively build the tree by selecting the best features at each node.

❖ Training the Decision Tree

- Train the decision tree classifier using the training data.

❖ Testing and Evaluation

- Use the trained decision tree to classify the test data.
- Evaluate the model performance using metrics such as accuracy, precision, recall, and F1-score.
- Visualize the decision tree (if possible) using a library matplotlib.

❖ Analysis and Conclusion

- Discuss the results, including any observations from the decision tree

structure. ➤ Reflect on the performance of the ID3 algorithm for the chosen dataset.